

ECO101

Introduction to Microeconomics

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Credit: Cengage 2014

Week 6 – Welfare & Efficiency

- *How do taxes affect market outcomes? How do the effects depend on whether the tax is imposed on buyers or sellers?*
- *What is the incidence of a tax? What determines the incidence?*

Taxes

The govt levies taxes on many goods & services to raise revenue to pay for national defense, public schools, etc.

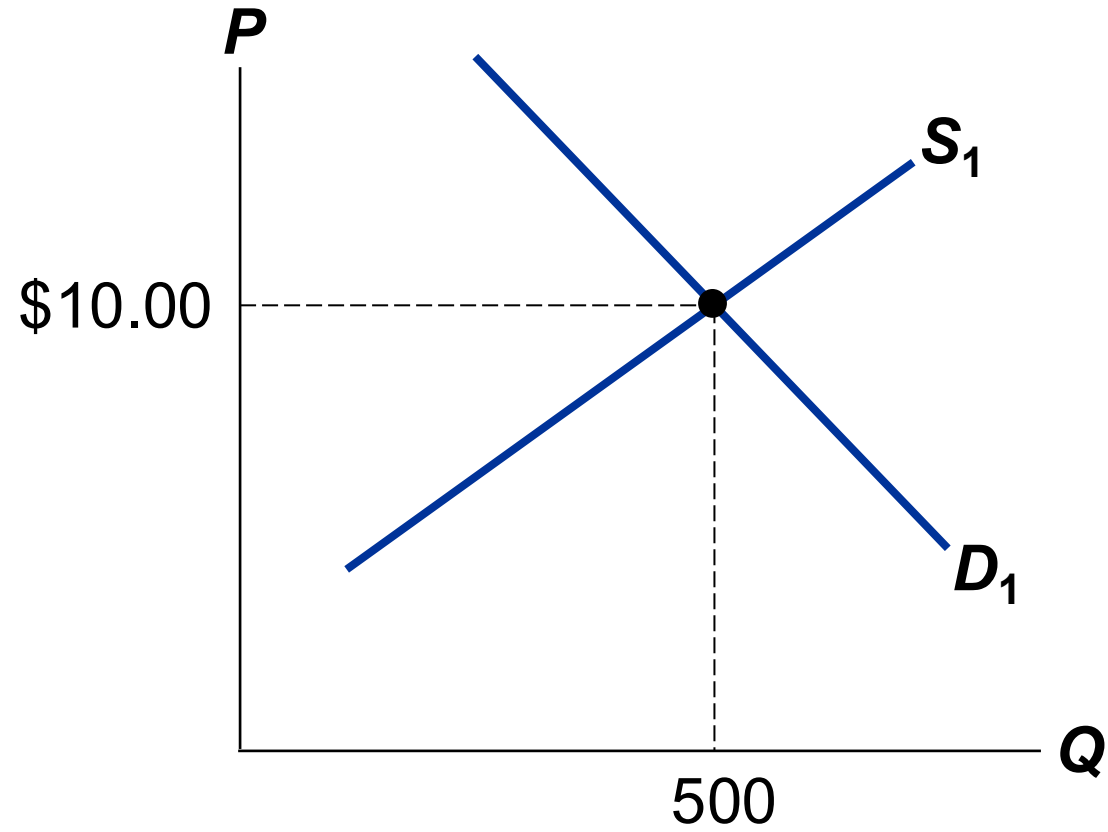
The govt can make buyers or sellers pay the tax.

The tax can be a % of the good's price, or a specific amount for each unit sold.

For simplicity, we analyze per-unit taxes only.

EXAMPLE 3: The Market for Pizza

Eq'm
w/o tax



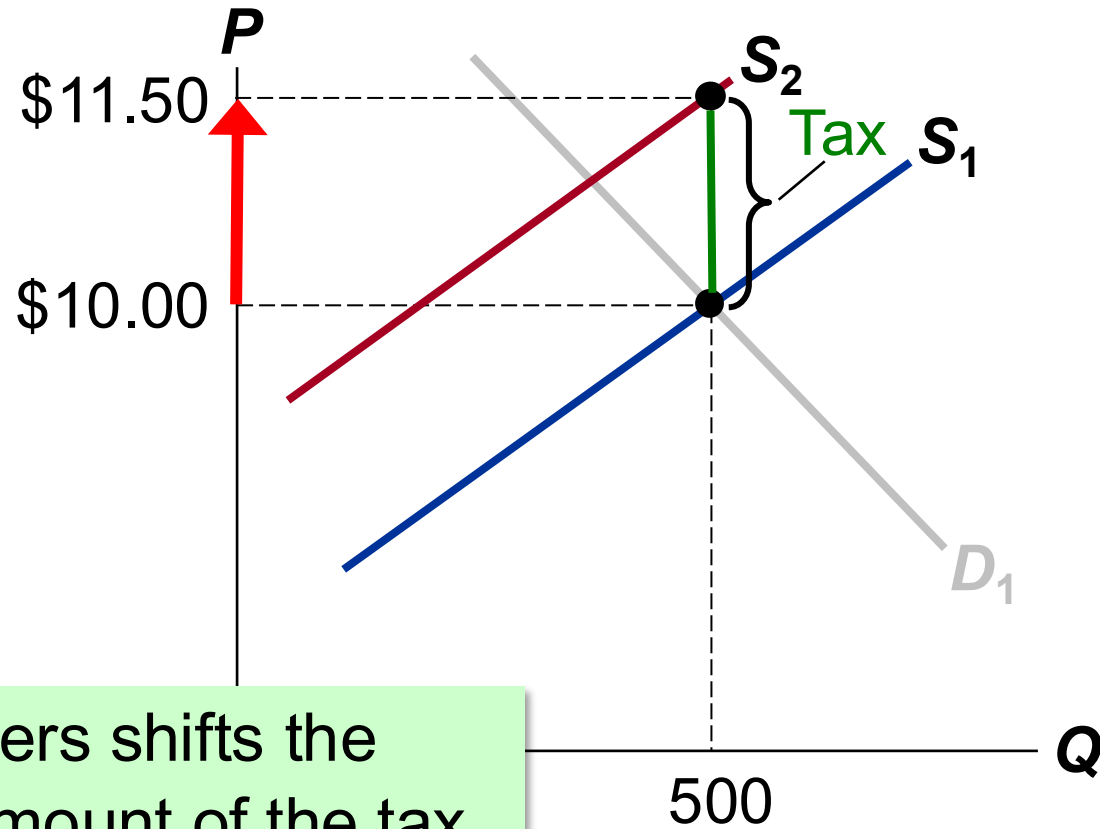
A Tax on Sellers

The tax effectively raises sellers' costs by \$1.50 per pizza.

Sellers will supply 500 pizzas only if P rises to \$11.50, to compensate for this cost increase.

Hence, a tax on sellers shifts the S curve up by the amount of the tax.

Effects of a \$1.50 per unit tax on sellers



A Tax on Sellers

New eq'm:

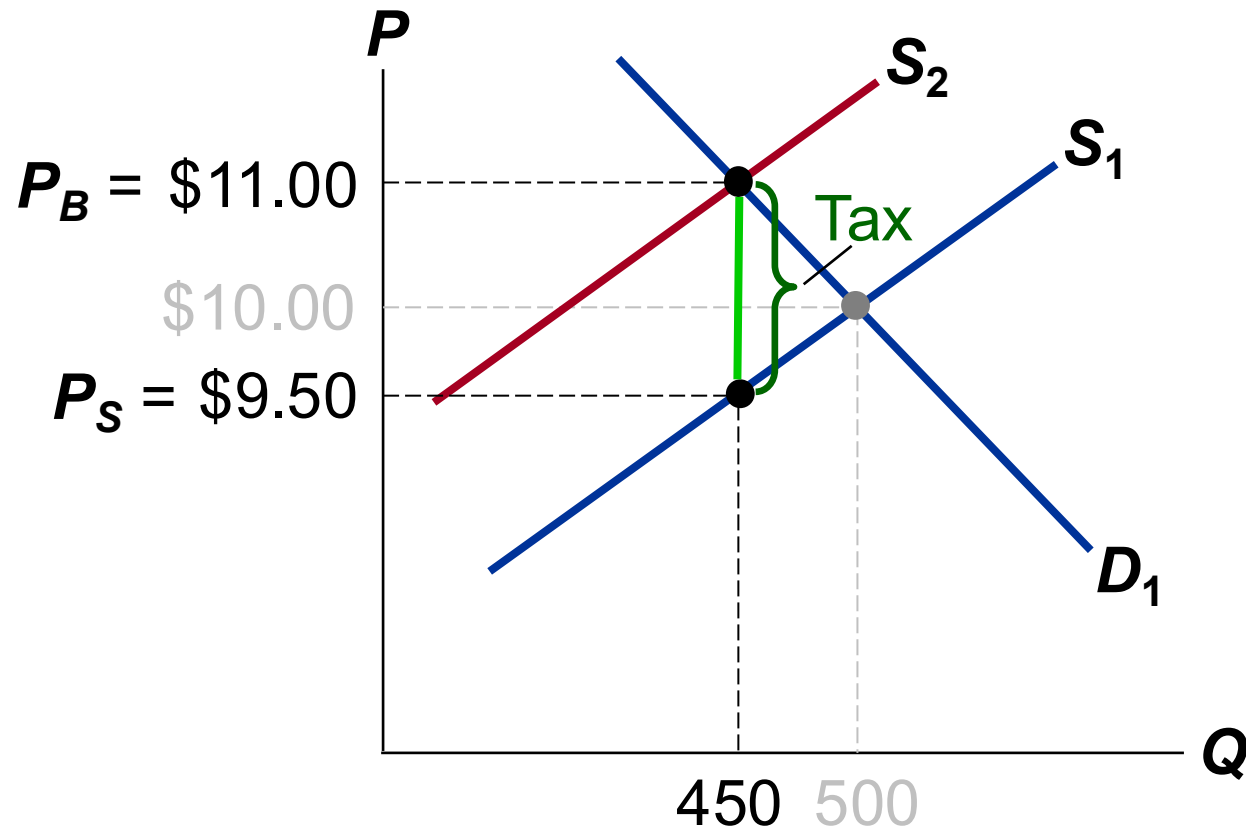
$Q = 450$

Buyers pay
 $P_B = \$11.00$

Sellers
receive
 $P_S = \$9.50$

Difference
between them
 $= \$1.50 = \text{tax}$

Effects of a \$1.50 per
unit tax on sellers



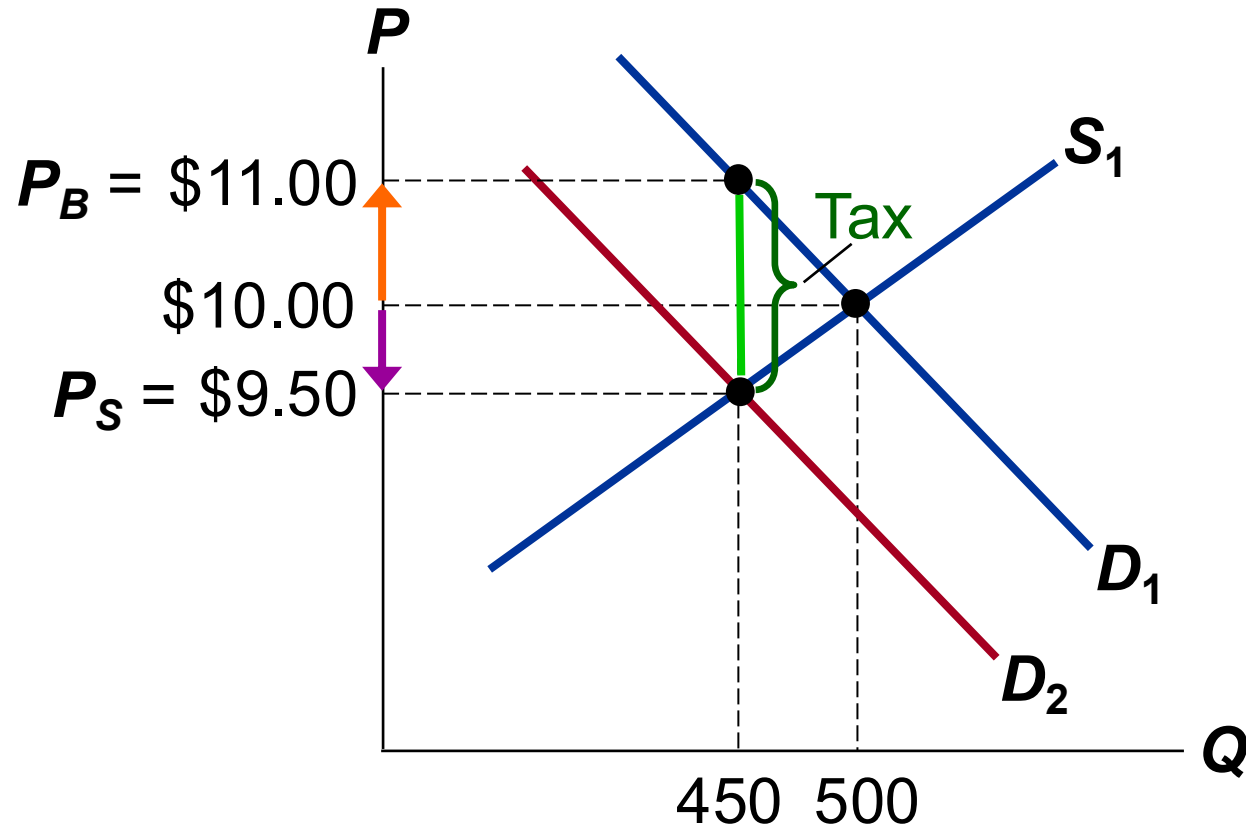
The Incidence of a Tax:

how the burden of a tax is shared among market participants

In our example,

buyers pay
\$1.00 more,

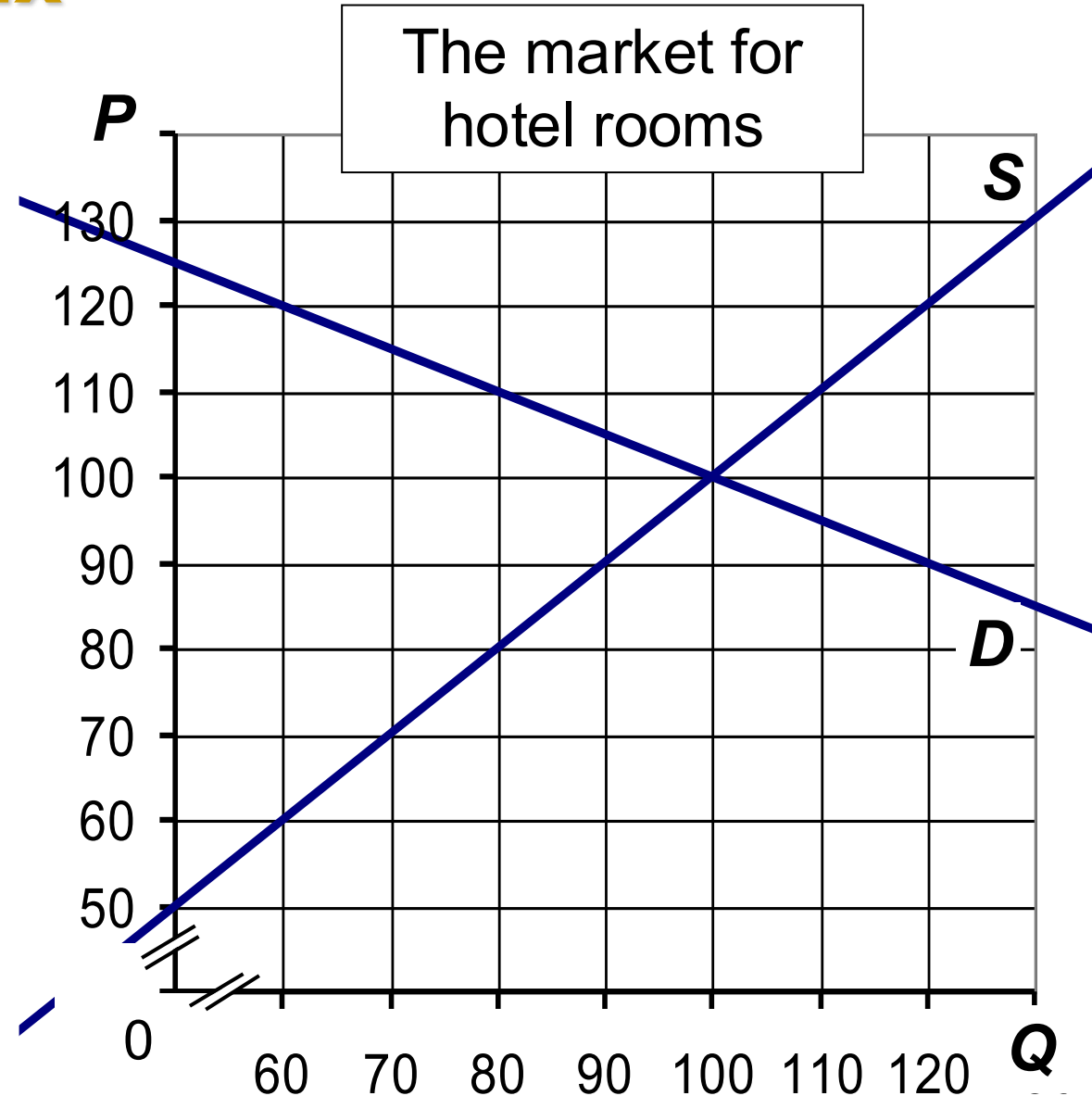
sellers get
\$0.50 less.



Effects of a tax

Suppose govt imposes a tax on buyers of \$30 per room.

Find new Q , P_B , P_S , and incidence of tax.



Effects of a tax

$$Q = 80$$

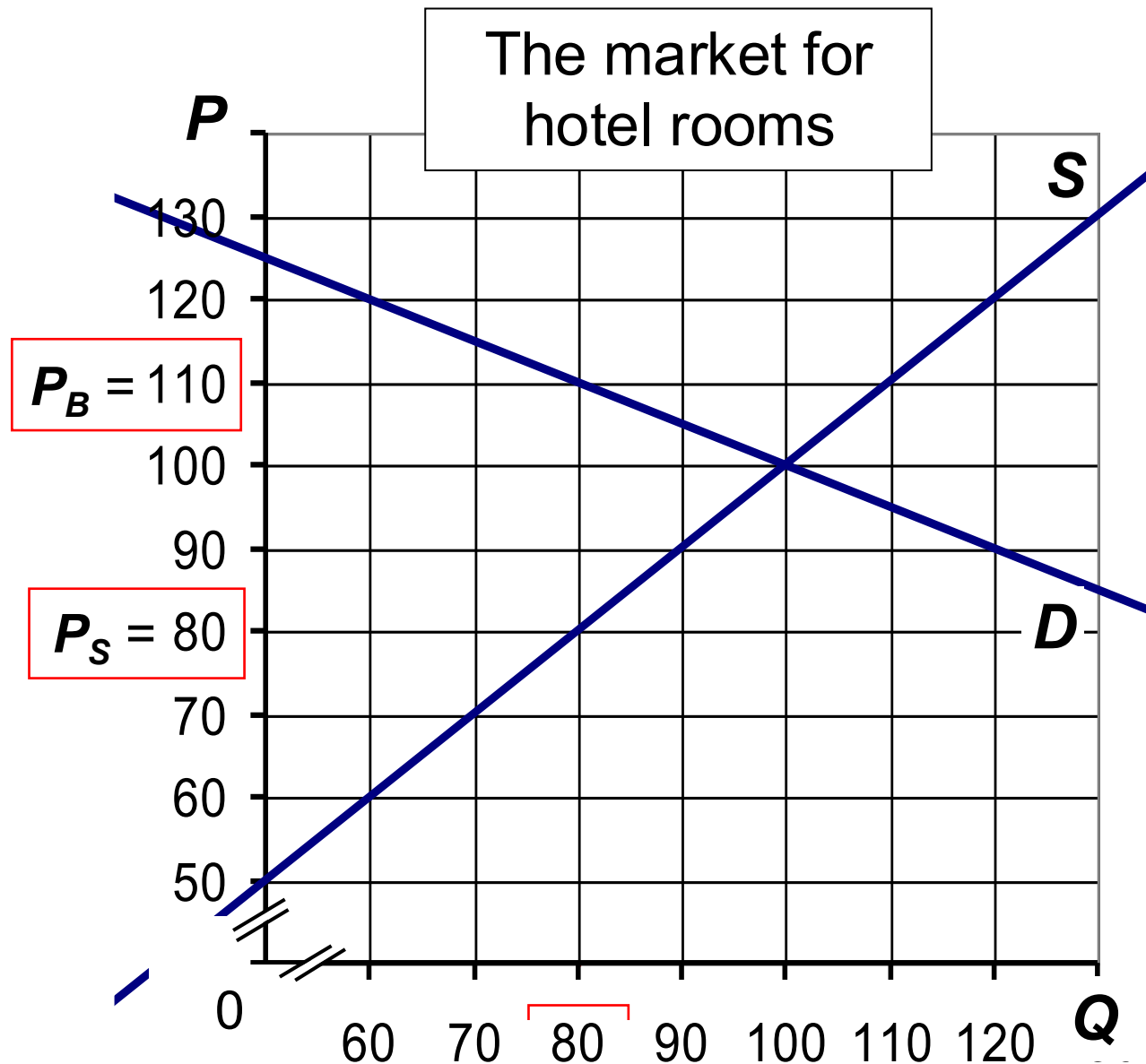
$$P_B = \$110$$

$$P_S = \$80$$

Incidence

buyers: \$10

sellers: \$20



The Effects of a Tax

Eq'm with no tax:

Price = P_E

Quantity = Q_E

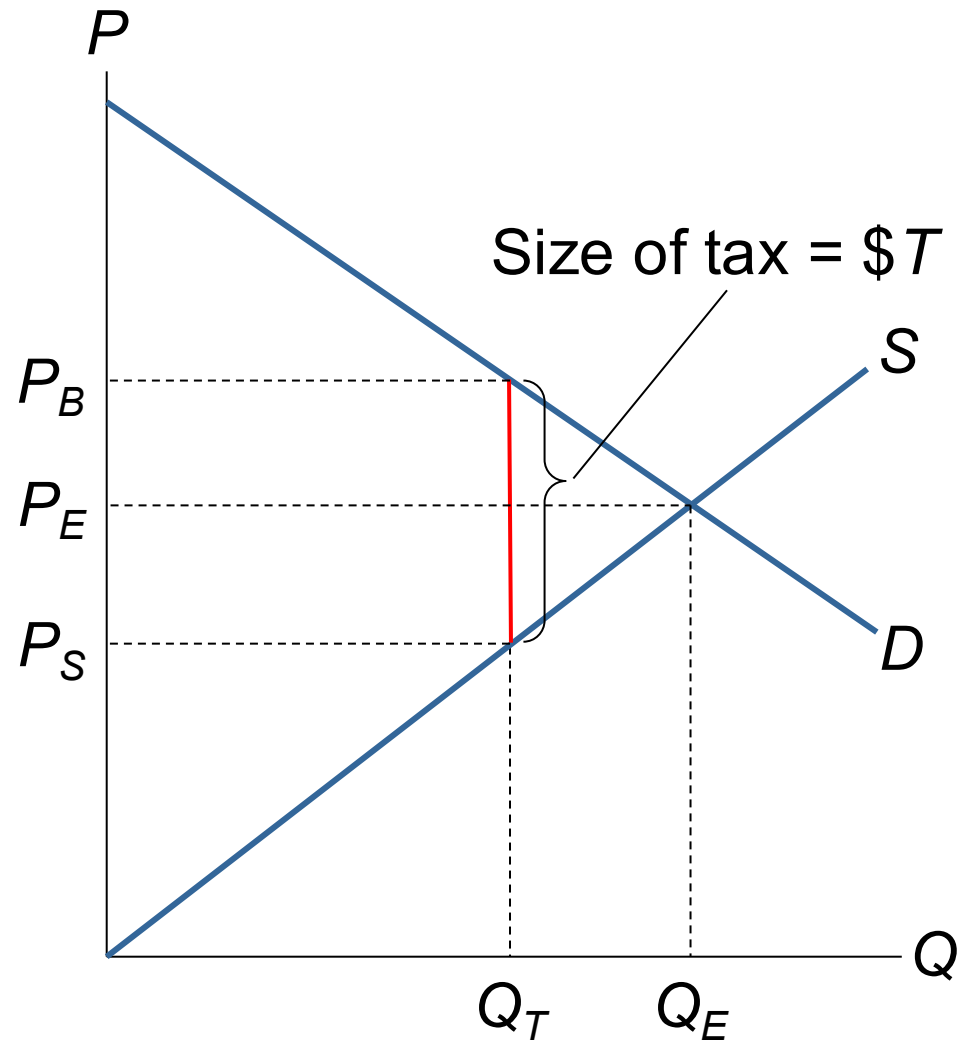
Eq'm with

tax = $\$T$ per unit:

Buyers pay P_B

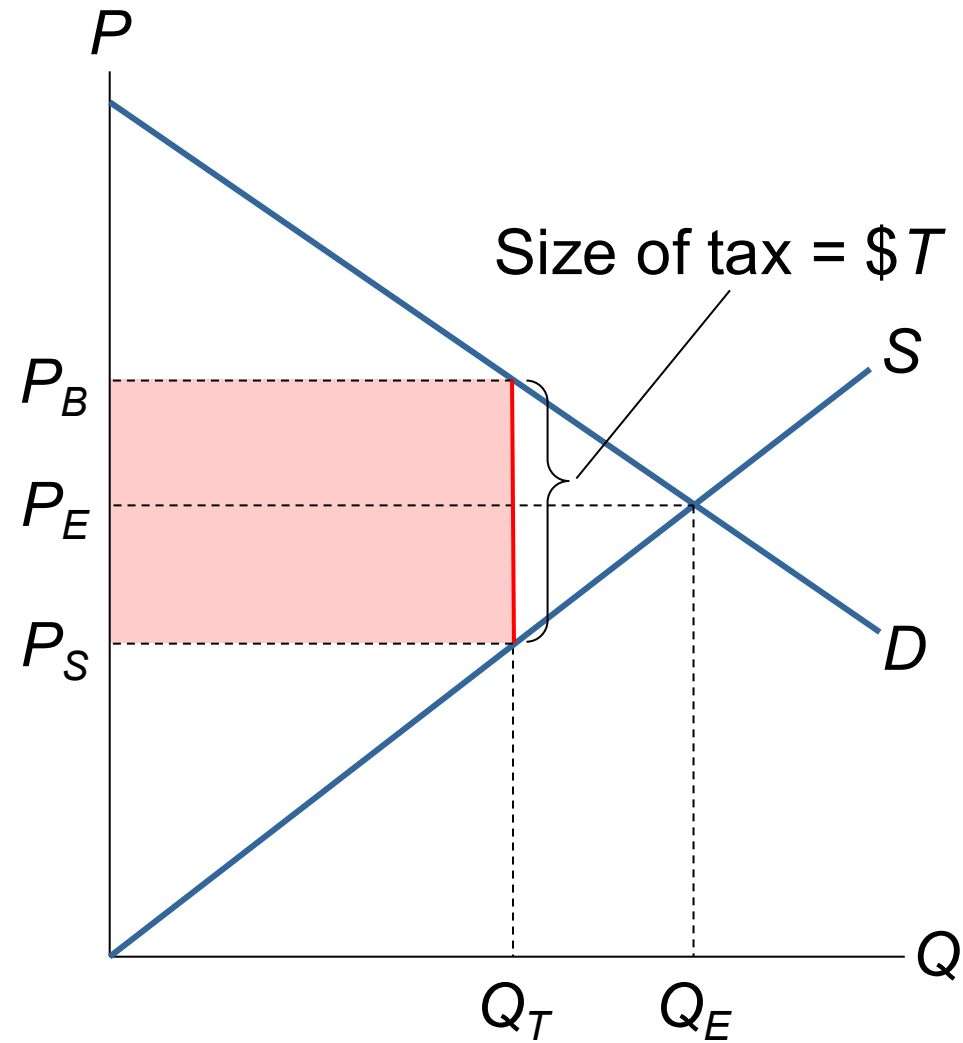
Sellers receive P_S

Quantity = Q_T



The Effects of a Tax

Revenue from tax:
 $\$T \times Q_T$



The Effects of a Tax

- Next, we apply welfare economics (the study of how the allocation of resources affects economic well-being) to measure the gains and losses from a tax.
- We determine consumer surplus (CS), producer surplus (PS), tax revenue, and total surplus with and without the tax.
- Tax revenue can fund beneficial services (e.g., education, roads, police), so we include it in total surplus.

The Effects of a Tax

Without a tax,

$$CS = A + B + C$$

$$PS = D + E + F$$

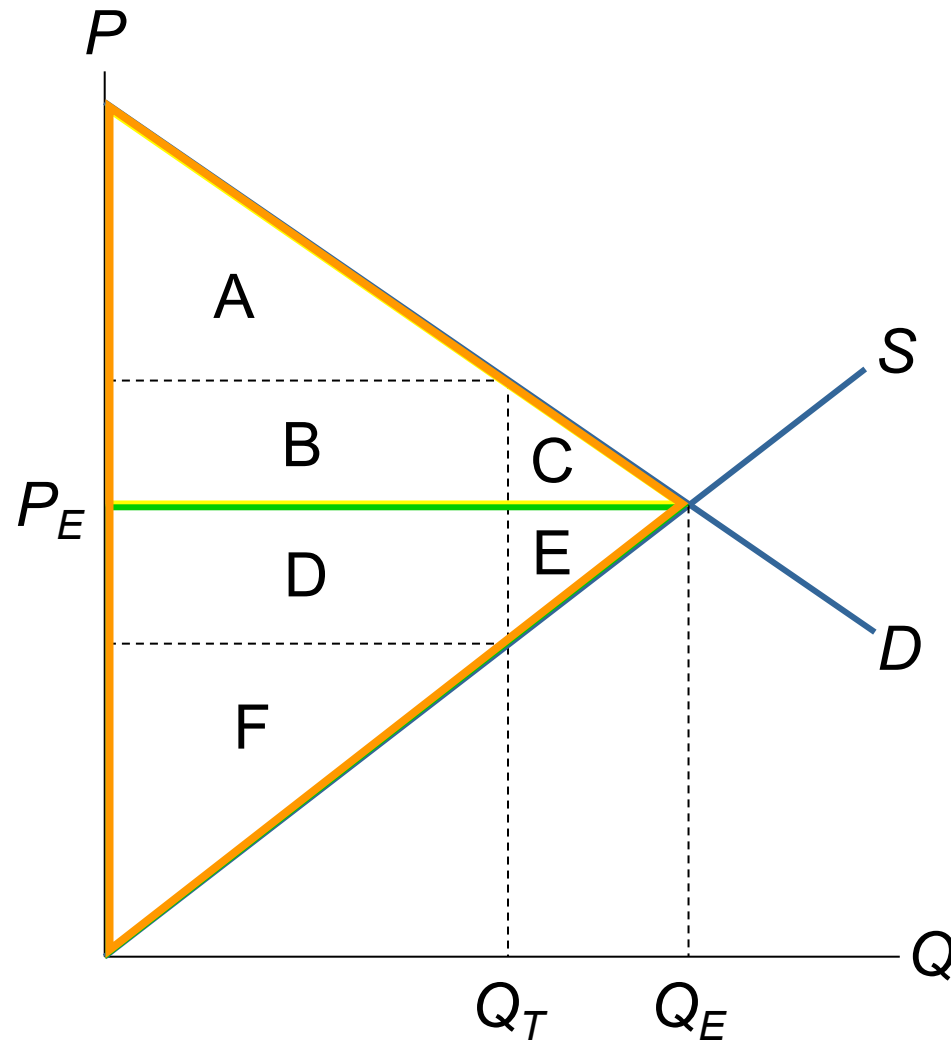
$$\text{Tax revenue} = 0$$

Total surplus

$$= CS + PS$$

$$= A + B + C$$

$$+ D + E + F$$



The Effects of a Tax

With the tax,

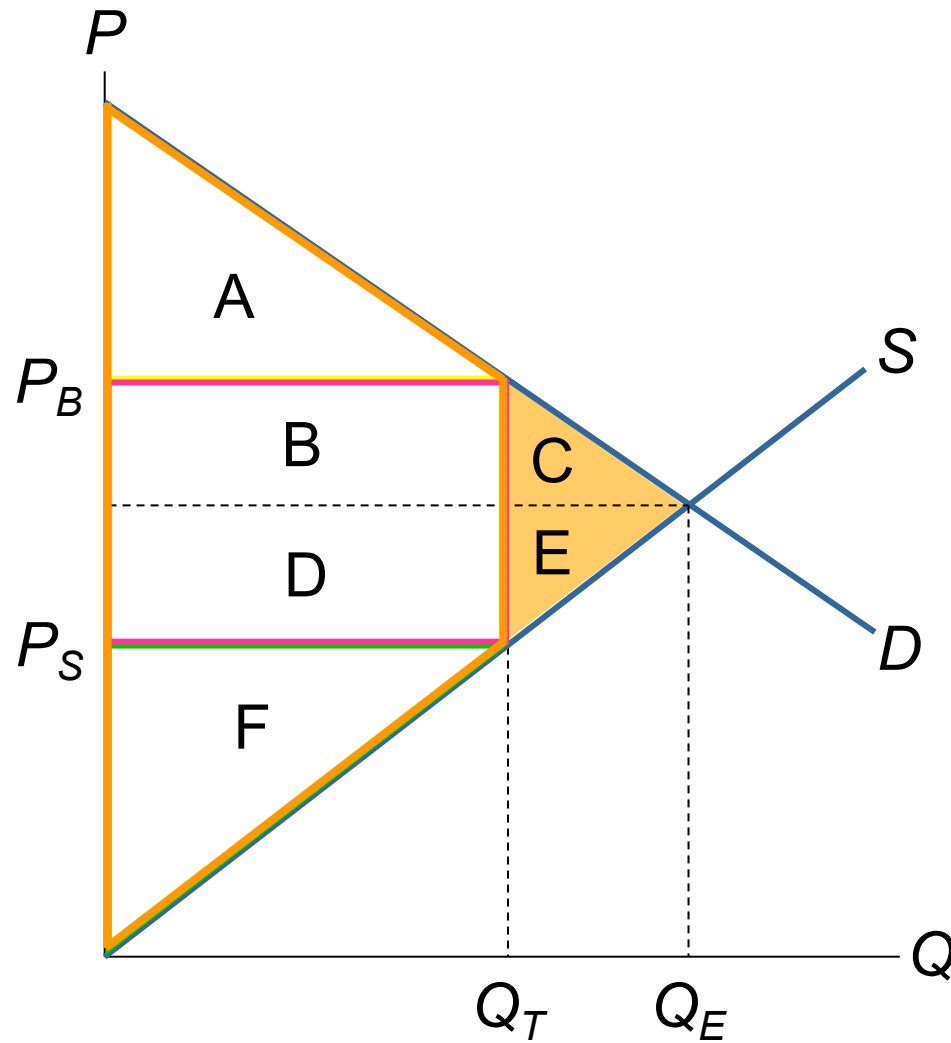
$$CS = A$$

$$PS = F$$

$$\begin{aligned}\text{Tax revenue} \\ &= B + D\end{aligned}$$

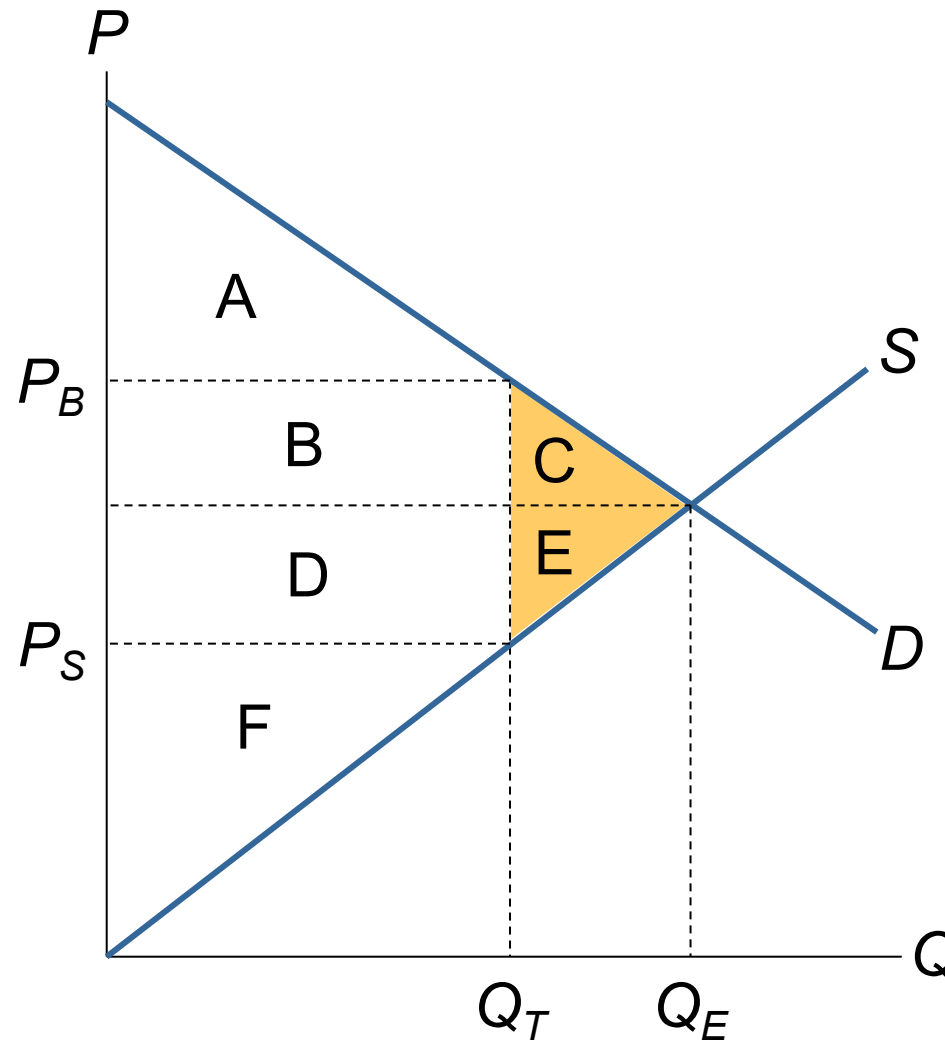
$$\begin{aligned}\text{Total surplus} \\ &= A + B \\ &\quad + D + F\end{aligned}$$

The tax reduces
total surplus by
 $C + E$



The Effects of a Tax

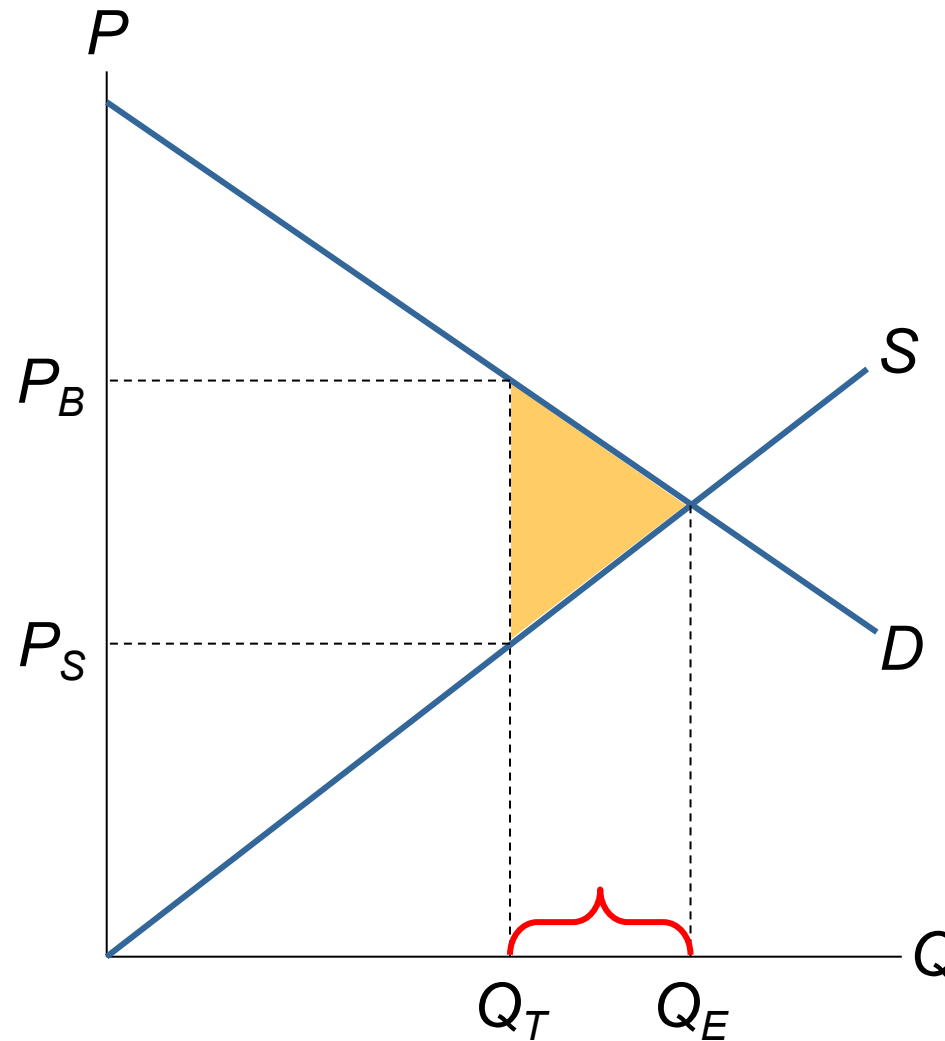
C + E is called the **deadweight loss** (DWL) of the tax, the fall in total surplus that results from a market distortion, such as a tax.



About the Deadweight Loss

Because of the tax, the units between Q_T and Q_E are not sold.

The value of these units to buyers is greater than the cost of producing them, so the tax prevents some mutually beneficial trades.



Summary

- A tax on a good places a wedge between the price buyers pay and the price sellers receive, and causes the eq'm quantity to fall, whether the tax is imposed on buyers or sellers.

- The incidence of a tax is the division of the burden of the tax between buyers and sellers, and does not depend on whether the tax is imposed on buyers or sellers.

Summary

- A tax on a good reduces the welfare of buyers and sellers. This welfare loss usually exceeds the revenue the tax raises for the govt.
- The fall in total surplus (consumer surplus, producer surplus, and tax revenue) is called the **deadweight loss (DWL)** of the tax.
- A tax has a DWL because it causes consumers to buy less and producers to sell less, thus shrinking the market below the level that maximizes total surplus.